JUNIOR LEVEL	PACKET TRACER	LAB 02

Lab 02

VLAN Configuration & Trunk Links

Segment a network with VLANs and configure 802.1Q trunks between switches

VLAN Creation	Access Ports	802.1Q Trunking	VLAN Verification	DTP Concepts

CCNA Topic	2.1 – Configure and verify VLANs (normal range)
Exam Objective	Configure and verify interswitch connectivity
Difficulty	Junior
Simulator	Cisco Packet Tracer 8.x
Devices	2x Cisco 2960 Switches + 4x PCs
Date Completed	2026-02-18

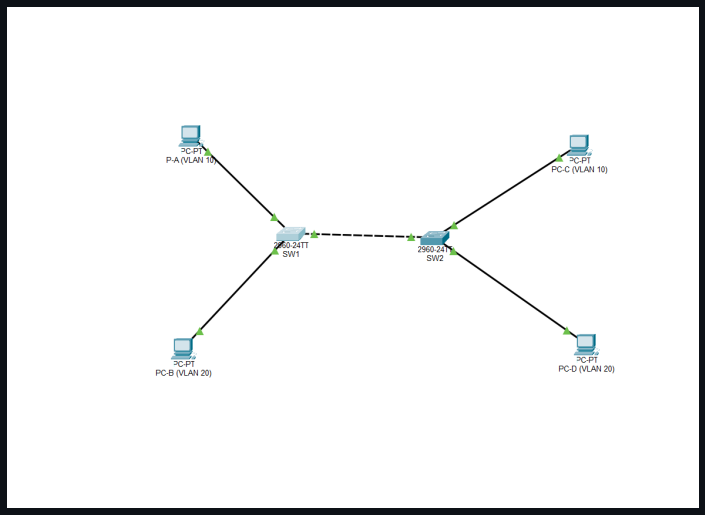


Figure 1 — Completed Packet Tracer topology: SW1 ↔ SW2 trunk (dashed line), VLAN 10 and VLAN 20 PCs

Lab Objective

VLANs are how every enterprise network is segmented. Without VLANs every device shares one broadcast domain — a security and performance problem. In this lab we created VLANs 10, 20, and 99 on two switches, assigned access ports, configured an 802.1Q trunk, and verified that VLAN traffic is correctly isolated. This is the most heavily tested topic on the CCNA exam.

Network Topology & Device Table

Device	Model	Hostname	VLAN	IP Address
Switch 1	2960	SW1	—	No IP this lab
Switch 2	2960	SW2	—	No IP this lab
PC-A	PC-PT	P-A (VLAN 10)	10	192.168.10.10/24
PC-B	PC-PT	PC-B (VLAN 20)	20	192.168.20.10/24
PC-C	PC-PT	PC-C (VLAN 10)	10	192.168.10.11/24
PC-D	PC-PT	PC-D (VLAN 20)	20	192.168.20.11/24

VLAN Table

VLAN ID	Name	Purpose	Subnet
10	SALES	PC-A and PC-C traffic	192.168.10.0/24
20	ENGINEERING	PC-B and PC-D traffic	192.168.20.0/24
99	MANAGEMENT	Native VLAN (security — no hosts)	N/A

Access vs Trunk Port Concepts

Port Type	Connects To	Carries	IOS Command
Access	End devices (PCs, printers)	One VLAN — untagged	switchport mode access
Trunk	Other switches	Multiple VLANs — 802.1Q tagged	switchport mode trunk

Step 1 — SW1: VLAN Creation & Access Ports

VLANs are created globally first, then assigned to ports. VLANs must be created on every switch that needs to carry that traffic — they do not propagate automatically without VTP.

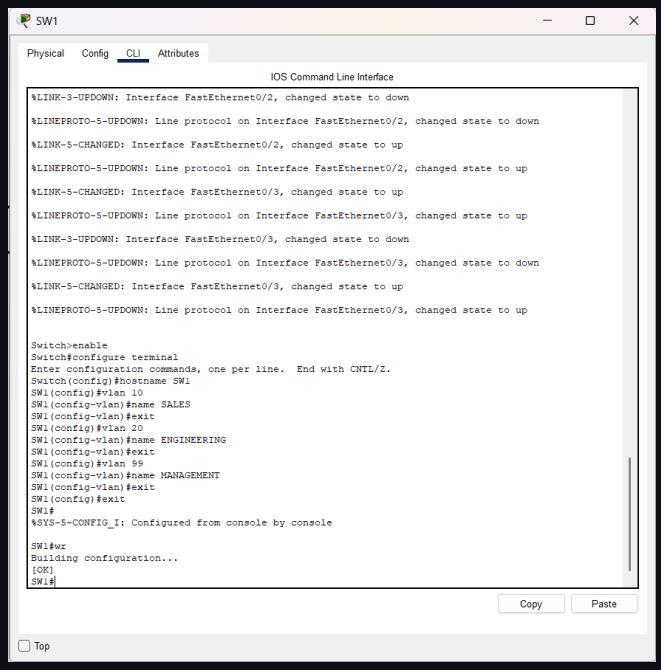


Figure 2 — SW1 CLI: hostname set, VLANs 10 (SALES), 20 (ENGINEERING), 99 (MANAGEMENT) created and saved

SW1 VLAN creation commands:

```
enable configure terminal hostname SW1 vlan 10 name SALES exit vlan 20 name ENGINEERING exit vlan 99 name MANAGEMENT exit wr
```

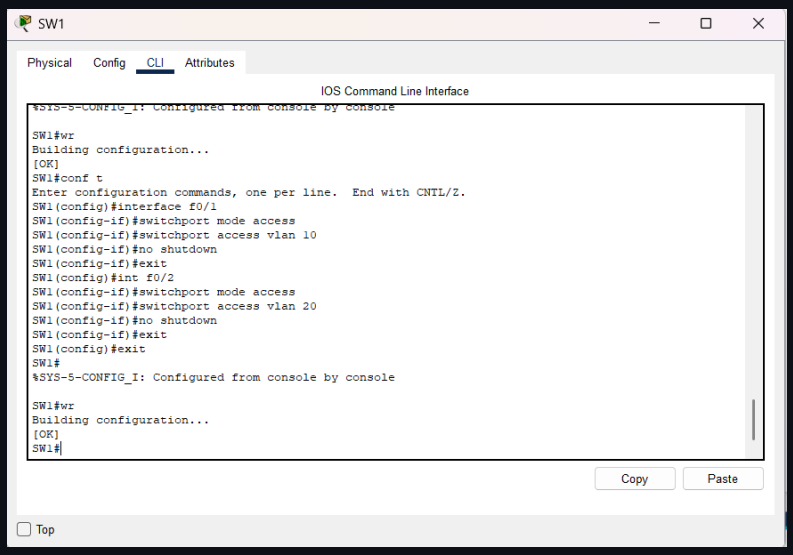


Figure 3 — SW1 CLI: Fa0/1 assigned to VLAN 10 (access), Fa0/2 assigned to VLAN 20 (access)

SW1 access port commands:

```
configure terminal interface FastEthernet0/1 switchport mode access switchport access vlan 10 no shutdown exit interface FastEthernet0/2 switchport mode access switchport access vlan 20 no shutdown exit wr
```

Step 2 — SW2: VLAN Creation & Access Ports

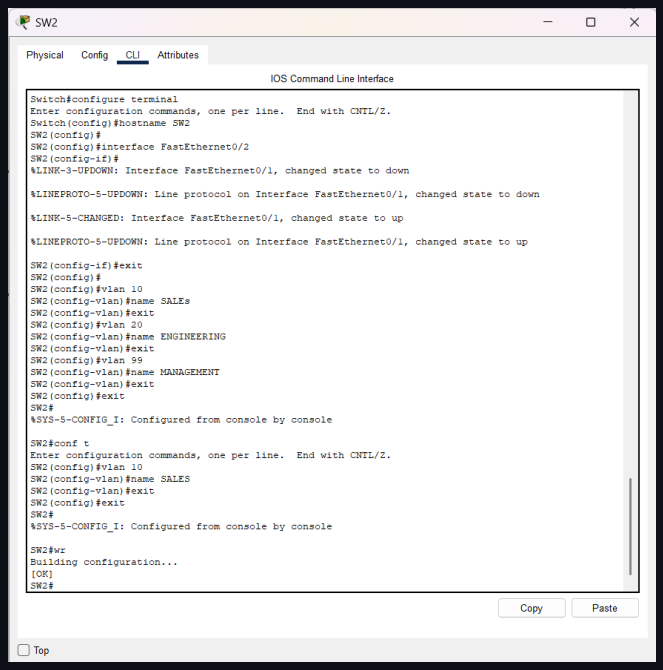


Figure 4 — SW2 CLI: VLAN creation (note: VLAN 10 initially named "SALEs" — caught and corrected)

■■ ISSUE FLAGGED: SW2 VLAN Name Typo

VLAN 10 was initially named "SALEs" (wrong capitalisation). IOS accepted it without error. A second config session corrected it to "SALES". VLAN

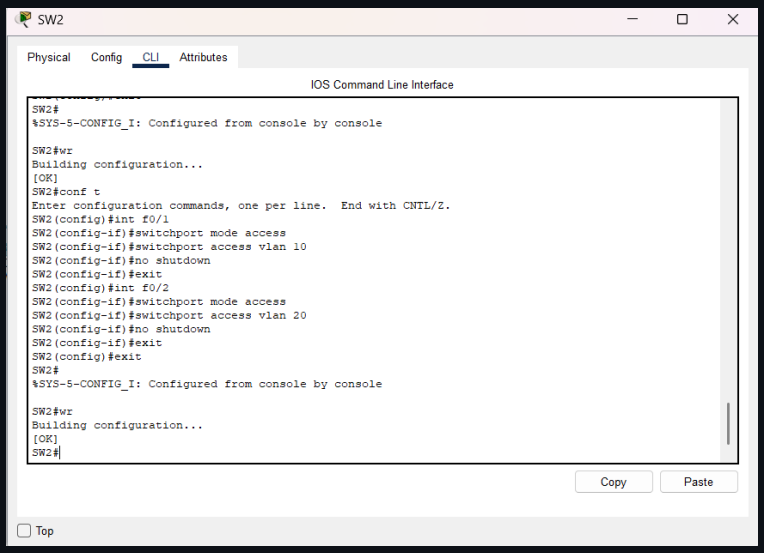


Figure 5 — SW2 CLI: Fa0/1 → VLAN 10 (access), Fa0/2 → VLAN 20 (access), saved

SW2 access port commands:

```
configure terminal interface FastEthernet0/1 switchport mode access switchport access vlan 10 no shutdown
exit interface FastEthernet0/2 switchport mode access switchport access vlan 20 no shutdown exit wr
```

Step 3 — Trunk Link Configuration

The trunk carries all three VLANs (10, 20, 99) between SW1 and SW2 using 802.1Q encapsulation. Native VLAN is set to 99 (MANAGEMENT) — an unused VLAN — which is a security best practice to prevent VLAN hopping attacks.

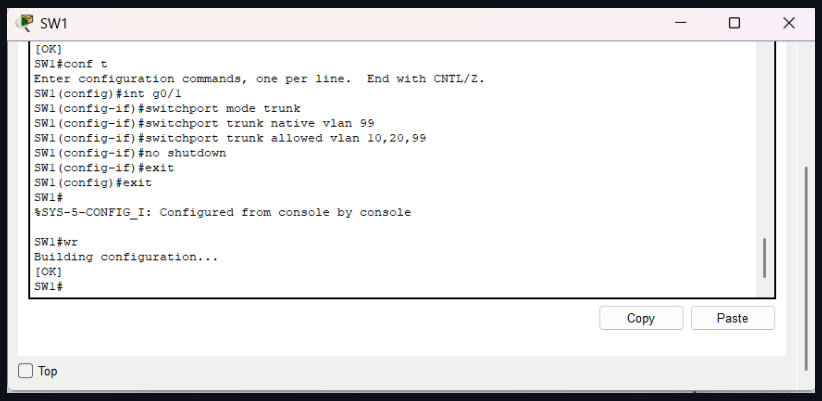


Figure 6 — SW1 Gi0/1: trunk mode, native VLAN 99, allowed VLANs 10,20,99 configured and saved

SW1 trunk commands:

```
configure terminal interface GigabitEthernet0/1 switchport mode trunk switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,99 no shutdown exit wr
```

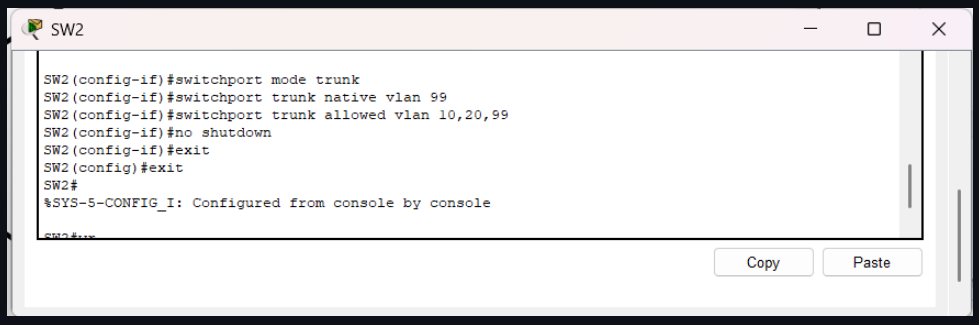


Figure 7 — SW2 Gi0/1: matching trunk config — mode trunk, native VLAN 99, allowed 10,20,99

SW2 trunk commands:

```
configure terminal interface GigabitEthernet0/1 switchport mode trunk switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,99 no shutdown exit wr
```

■■ TROUBLESHOOTING: Native VLAN Mismatch

SW2's trunk was configured with native VLAN 99 before SW1 was updated. This caused CDP and STP errors: %CDP-4-NATIVE_VLAN_MISMAT

Step 4 — PC IP Configuration

All four PCs are assigned static IPs. No default gateway is configured — there is no router in this lab. Inter-VLAN routing requires a router and is covered in Lab 03.

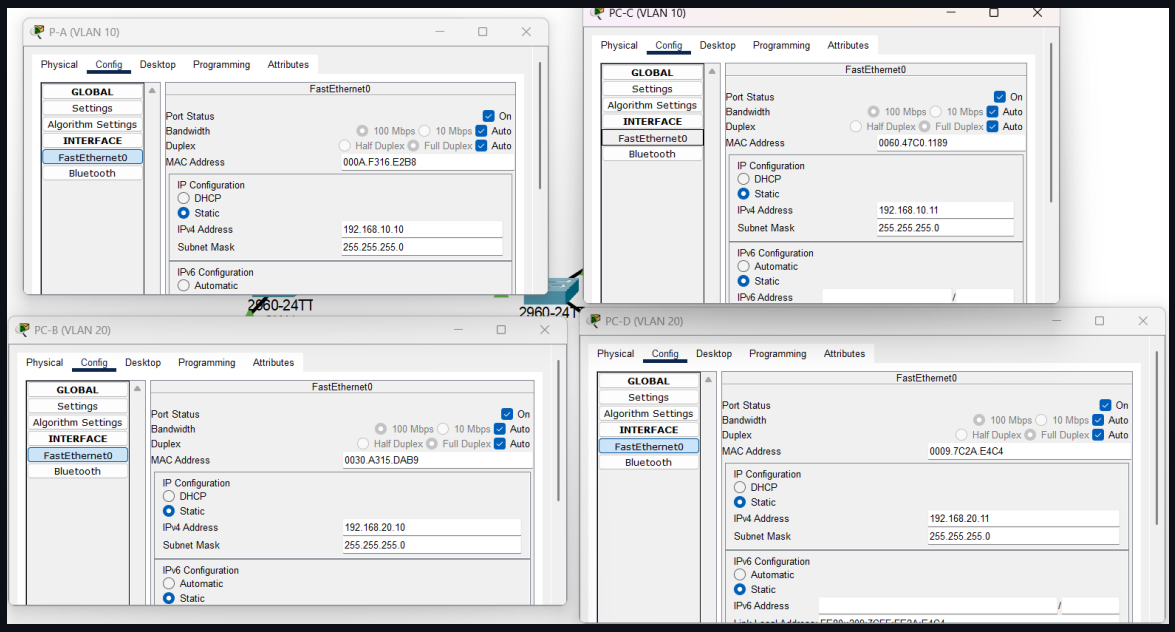


Figure 8 — All four PCs with static IPs: PC-A (10.10), PC-B (20.10), PC-C (10.11), PC-D (20.11)

PC	IP Address	Subnet Mask	Default Gateway	VLAN
PC-A	192.168.10.10	255.255.255.0	None	10 — SALES
PC-B	192.168.20.10	255.255.255.0	None	20 — ENGINEERING
PC-C	192.168.10.11	255.255.255.0	None	10 — SALES
PC-D	192.168.20.11	255.255.255.0	None	20 — ENGINEERING

Step 5 — VLAN Isolation Verification

VLAN isolation is proven by two types of test: same-VLAN pings across the trunk (must succeed), and cross-VLAN pings (must fail with 100% loss). Both tests were performed from PC-A and PC-B.

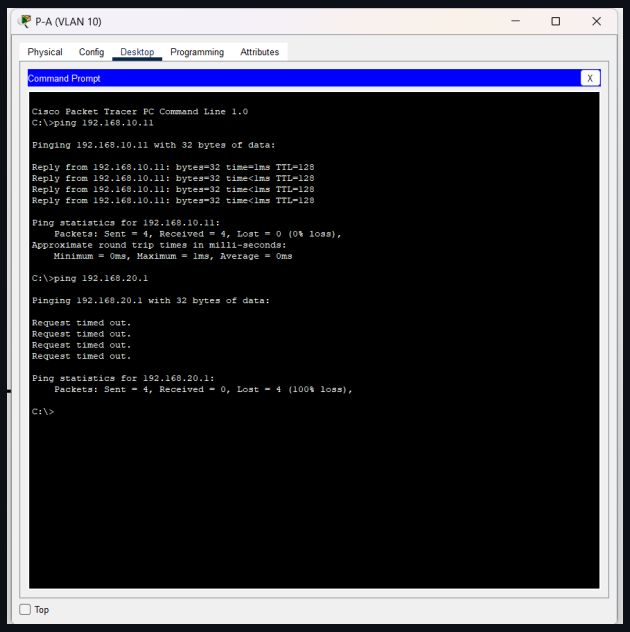


Figure 9 — PC-A: ping PC-C (same VLAN 10, across trunk) → SUCCESS. ping 192.168.20.x (VLAN 20) → TIMEOUT

> **■ Note: Test Accuracy**
>
> In Figure 9, PC-A pinged 192.168.20.1 instead of PC-B at 192.168.20.10. The ping timed out, but this was because 192.168.20.1 does not exist —

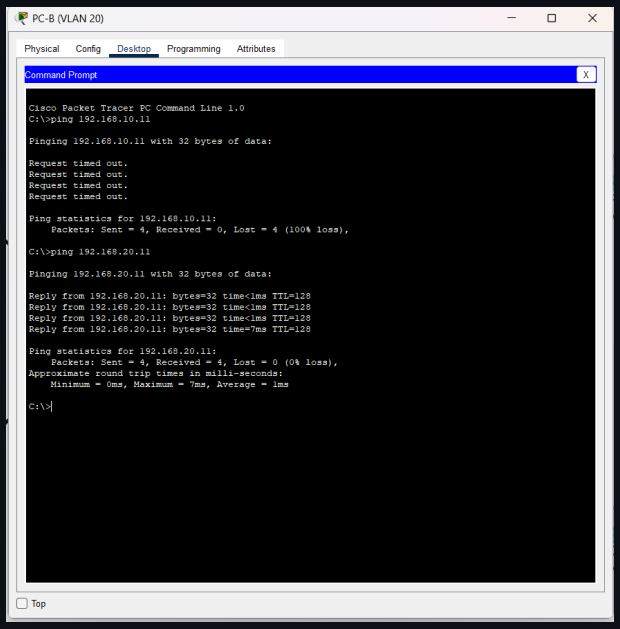


Figure 10 — PC-B: ping PC-C (VLAN 10) → 100% LOSS (isolation confirmed). ping PC-D (VLAN 20) → SUCCESS

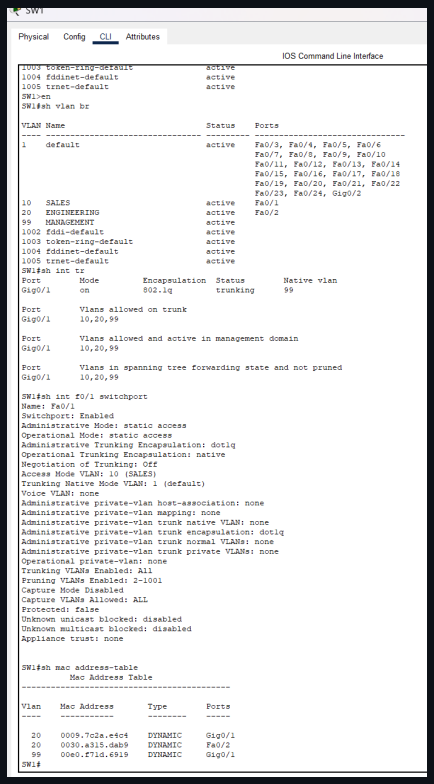
Ping Test Results Summary

Source	Target	Target IP	VLAN Relationship	Result

PC-A	PC-C	192.168.10.11	Same VLAN 10 — across trunk	4/4 — 0% loss ✓
PC-A	PC-B	192.168.20.10	Cross VLAN 10→20	100% loss ✓
PC-B	PC-C	192.168.10.11	Cross VLAN 20→10	100% loss ✓
PC-B	PC-D	192.168.20.11	Same VLAN 20 — across trunk	4/4 — 0% loss ✓

Step 6 — Verification Show Commands

Three verification commands were run on SW1 to confirm the complete configuration: show vlan brief, show interfaces trunk, and show interfaces switchport. The MAC address table was also captured.


SW1
Physical
Config
CLI
Attributes
IOS Command Line Interface
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
SW1>en
SW1#sh vlan br
VLAN Name Status Ports
---- ---|---|---|

Key Takeaways

VLAN Isolation is Layer 2 Only

Devices in different VLANs cannot communicate without a router — no matter how they are cabled. This is the fundamental security purpose of VLANs.

VLANs Must Exist on Every Switch

Without VTP, VLANs must be created manually on each switch that carries that traffic. A switch that does not know a VLAN will drop those frames.

Native VLAN Must Match on Both Trunk Ends

A mismatch causes CDP errors and STP blocks the port. Always set native VLAN to an unused VLAN (99 here) to prevent VLAN hopping security attacks.

Access Ports vs Trunk Ports

Access ports carry one untagged VLAN — for end devices. Trunk ports carry multiple VLANs using 802.1Q tags — for switch-to-switch links.

show vlan brief is Your First Check

When VLAN traffic is not working, always run show vlan brief first. Confirm the VLAN exists and the correct port is assigned.

show interfaces trunk is Critical

Confirms mode, encapsulation, native VLAN, and allowed VLANs. All three must be correct on both ends before inter-switch VLAN traffic will pass.

No Router = No Inter-VLAN Traffic

VLAN isolation is intentional. To allow communication between VLANs you need a router (or Layer 3 switch). This is covered in Lab 03: Inter-VLAN Routing.

Lab Validation Checklist

- ✓ VLANs 10 (SALES), 20 (ENGINEERING), 99 (MANAGEMENT) created on SW1
- ✓ VLANs 10 (SALES), 20 (ENGINEERING), 99 (MANAGEMENT) created on SW2
- ✓ SW1 Fa0/1 → access VLAN 10 | SW1 Fa0/2 → access VLAN 20
- ✓ SW2 Fa0/1 → access VLAN 10 | SW2 Fa0/2 → access VLAN 20
- ✓ Trunk link: SW1 Gi0/1 ↔ SW2 Gi0/1 — mode trunk, 802.1Q, native VLAN 99
- ✓ Trunk allowed VLANs: 10, 20, 99 on both ends
- ✓ Native VLAN mismatch resolved — no CDP or STP errors
- ✓ PC-A (VLAN 10) → PC-C (VLAN 10) across trunk: 0% loss ■
- ✓ PC-A (VLAN 10) → VLAN 20 addresses: 100% loss — isolation confirmed ■
- ✓ PC-B (VLAN 20) → PC-D (VLAN 20) across trunk: 0% loss ■
- ✓ PC-B (VLAN 20) → PC-C (VLAN 10): 100% loss — isolation confirmed ■
- ✓ show vlan brief confirms port-to-VLAN assignments
- ✓ show interfaces trunk confirms 802.1Q, native 99, allowed 10,20,99
- ✓ show interfaces switchport confirms access mode and VLAN assignment

Lab 02 Complete — VLANs are the #1 most tested CCNA topic. Master them.

Next: Lab 03 — Inter-VLAN Routing (Router-on-a-Stick)